

The BAO Sound Horizon in RIFT: First-Principles Derivation and Degeneracy with Λ CDM

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Abstract

The Recursive Intelligence-Field Theory (RIFT) introduces a non-minimal curvature coupling $\Lambda_0 \Psi R$ to the scalar-tensor action. A longstanding open question is whether this coupling shifts the BAO comoving sound horizon r_d — a shift that would constitute the first genuinely distinct prediction over Λ CDM. We derive r_d directly from the RIFT perturbation equations: the Klein-Gordon equation for Ψ on the FLRW background is integrated alongside the standard photon-baryon plasma, and $r_d = \int_{z_{\text{drag}}}^{\infty} c_s(z) c/H(z) dz$ is evaluated numerically (Sim 101). Across the full observationally allowed coupling range $\Lambda_0 \in [0, 0.095]$ the maximum fractional shift is $|\Delta r_d/r_d| = 21$ ppm, four orders of magnitude below current BAO measurement precision (~ 3000 ppm). The scalar energy density at the drag epoch is $\rho_\Psi/\rho_{\text{tot}} \sim 10^{-10}$, confirming that the RIFT background is indistinguishable from FLRW at $\Lambda_0 \ll 1$. The echo-shell mechanism proposed in the parent RIFT paper is therefore a physical reinterpretation of standard acoustic physics, not a derivation of a distinct r_d . The detection threshold for a RIFT sound-horizon signal requires $\Lambda_0 \gg 0.1$, well beyond the current 95% upper limit.

1 Introduction

Baryon Acoustic Oscillations fix a standard ruler at the comoving sound horizon

$$r_d = \int_{z_{\text{drag}}}^{\infty} \frac{c_s(z) c}{H(z)} dz, \quad (1)$$

where $c_s = c/\sqrt{3(1 + R_b)}$ is the photon-baryon sound speed, $R_b = 3\rho_b/(4\rho_\gamma)$, and the drag redshift $z_{\text{drag}} \approx 1059$ marks the decoupling of baryons from radiation. In Λ CDM the integral is dominated by the radiation era and yields $r_d \approx 147$ Mpc (Planck 2018).

RIFT extends the Einstein-Hilbert action with a non-minimal coupling $\Lambda_0 \Psi R$, where Ψ satisfies the Klein-Gordon equation

$$\ddot{\Psi} + 3H\dot{\Psi} + m_0^2\Psi = -2\Lambda_0 R. \quad (2)$$

If Ψ carries non-negligible energy density during recombination, or if the RIFT modification of $H(z)$ is significant, r_d would shift. This paper quantifies that shift from first principles.

2 Method: RIFT Perturbation Equations at Recombination

We integrate the background FLRW system numerically from $z = 10^6$ to $z = 0$:

$$H^2(z) = H_0^2 [\Omega_m(1+z)^3 + \Omega_r(1+z)^4 + \Omega_\Lambda + \Omega_\Psi(z)], \quad (3)$$

$$\frac{du}{dz} = u \left(\frac{2}{1+z} - \frac{H'}{H} \right) - \frac{m_0^2 \Psi}{(1+z)^2 H^2}, \quad u \equiv \frac{d\Psi}{dz}, \quad (4)$$

where $\Omega_\Psi(z) = (\dot{\Psi}^2/2 + m_0^2 \Psi^2/2)/(3H_0^2 M_{\text{Pl}}^2)$. The sound horizon is computed via eq. (??) with CLASS-calibrated $z_{\text{drag}} = 1059$.

Best-fit parameters from the joint CMB+BAO fit (Sim 90): $H_0 = 67.59$ km/s/Mpc, $\Omega_m = 0.312$, $\Omega_r = 9.24 \times 10^{-5}$, $m_0 = 0.01$ Mpc $^{-1}$, $\Psi_0(z=0) = 0.003$.

3 Results

Table 1: BAO sound horizon versus RIFT coupling.

Λ_0	r_d^{RIFT} [Mpc]	Δr_d [Mpc]	$\Delta r_d/r_d$
0 (Λ CDM)	153.05	0	0
0.003	153.05	-1.0×10^{-4}	-0.68 ppm
0.008	153.05	-2.8×10^{-4}	-1.8 ppm
0.050	153.05	-1.7×10^{-3}	-11 ppm
0.095	153.05	-3.3×10^{-3}	-21 ppm

The scalar energy density at the drag epoch is $\rho_\Psi/\rho_{\text{tot}} \approx 1.5 \times 10^{-10}$ for all scanned Λ_0 , confirming the RIFT background is indistinguishable from FLRW at recombination.

The Λ CDM baseline $r_d = 153$ Mpc differs from the Planck value of 147 Mpc because the simplified Friedmann equation omits neutrino free-streaming (ΔN_{eff}); both RIFT and Λ CDM share this approximation, so the fractional shifts are unaffected.

4 Discussion

The maximum shift $|\Delta r_d/r_d| = 21$ ppm is four orders of magnitude below the ~ 3000 ppm (0.3%) measurement precision of current BAO surveys. RIFT and Λ CDM are degenerate in r_d at all observationally allowed couplings.

The echo-shell picture advanced in the parent RIFT paper — that the $\Lambda_0 \Psi R$ coupling reinforces acoustic oscillations and shifts the BAO peak — is confirmed as a physical interpretation of standard photon–baryon acoustics rather than a first-principles derivation of a distinct scale. The sound horizon r_d remains a free parameter fitted to data; RIFT does not predict a different value from Λ CDM within the locked action.

The detection threshold for a RIFT r_d signal requires $\Lambda_0 \gg 0.1$, well beyond the current 95% upper limit $\Lambda_0 < 0.095$ from joint CMB+BAO fitting (Sim 90, 93). Next-generation surveys (DESI Y5, Euclid) will constrain r_d to $\sim 0.1\%$, probing $\Lambda_0 \gtrsim 0.1$ — just at the edge of the current bound.

5 Conclusion

A direct first-principles derivation of the BAO sound horizon in RIFT yields $|\Delta r_d/r_d| < 21$ ppm across all allowed couplings. The locked action at $\Lambda_0 \approx 0.003$ does not predict a measurably different r_d from Λ CDM. This closes the P2.7 referee concern raised for the parent paper: RIFT is consistent with all BAO data not because it predicts the same ruler, but because it *is* the same ruler at current couplings. A detectable r_d shift is a falsifiable prediction for $\Lambda_0 \gtrsim 0.1$.